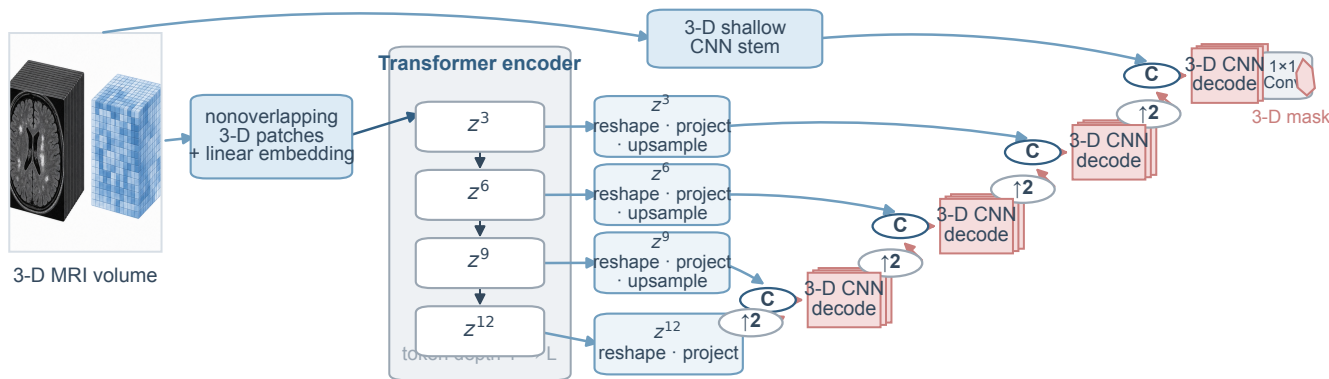


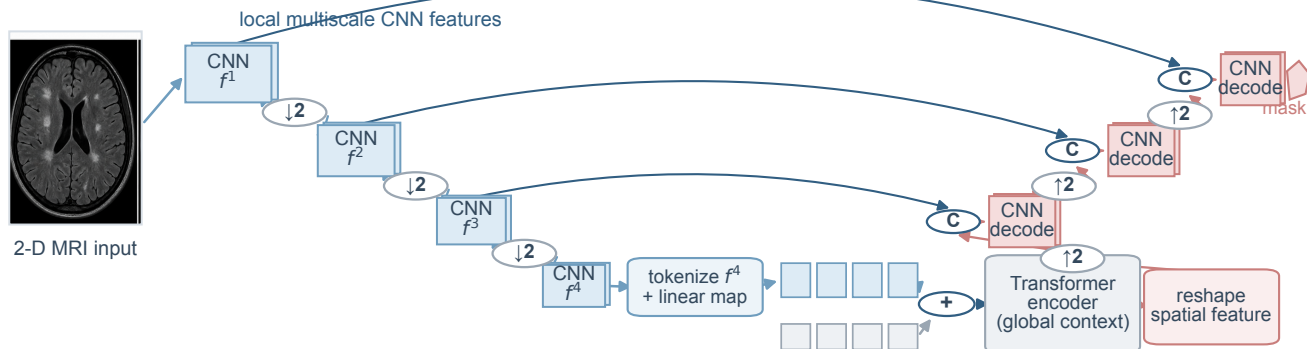
Transformer–decoder integration patterns in 3-D and hybrid segmentation

(a) UNETR: volumetric tokens and multidepth Transformer skips



Every hidden state is reshaped to a spatial tensor and projected before channel Concat
 z^{12} initializes decoding; $z^9 \rightarrow z^6 \rightarrow z^3 \rightarrow$ shallow CNN stem restore progressively finer detail

(b) Representative CNN–Transformer hybrid: TransUNet



CNN feature extraction precedes tokenization; retained $f^3/f^2/f^1$ skips restore local detail

C = channel Concat of a shape-aligned CNN skip and an upsampled decoder feature